SUMMIT

STEM CATEGORY

SLOT 1

STEM DIGICON

Computer Category

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INTRODUCTION

- STEM Digicon challenges students from Grades 8–11 to apply their digital literacy, logical thinking and problem-solving skills. Participants compete in both team-based and individual challenges, testing their knowledge of computer science, cybersecurity, number systems, AI-generated content and graphic designing.
- Through interactive rounds and a written test, students put their technical knowledge and practical skills to the test. This study guide outlines the key concepts and skills required to succeed in all rounds.

OVERVIEW

- The category consists of two main rounds: Squad Battle (Team Round) and Solo Showdown (Individual Round).
- Participants first compete in teams through four interactive stations, with the top-performing teams advancing to the second round.
- The final round is an individual written competition, where participants earn code points based on their performance.
- The competition carries a total of 100 code points, and the two participants with the highest overall scores are declared the winners.

CONTENT

STUDENTS REGISTERING AS STEM CARTEL NEED TO BE ABLE TO:

1. Computer Hardware and Architecture

- CPU, RAM, ROM and storage
- Input and output devices
- How a computer processes information

2. Number Systems and Data Representation

- Binary, decimal and hexadecimal
- Binary conversions
- Bits, bytes and data units
- ASCII and Unicode
- Image and sound representation

3. Cybersecurity and Digital Safety

- Malware and viruses
- Firewalls
- Phishing and online privacy

4. Control Sequences

- Students must memorize the list of Windows keyboard shortcuts below.

NO.	SHORTCUT	ACTION
1	**Ctrl + C**	Copy
2	**Ctrl + V**	Paste
3	**Ctrl + X**	Cut
4	**Ctrl + Z**	Undo
5	**Ctrl + Y**	Redo
6	**Ctrl + A**	Select all
7	**Ctrl + S**	Save
8	**Ctrl + P**	Print
9	**Ctrl + T**	New tab
10	**Ctrl + W**	Close tab
11	**Ctrl + Shift + T**	Reopen closed tab
12	**Ctrl + Tab**	Next tab
13	**Ctrl + Shift + Tab**	Previous tab
14	**Ctrl + L**	Focus address bar
15	**Ctrl + F**	Find on page
16	**Windows Key + E**	Open File Explorer
17	**Windows Key + D**	Display/hide the desktop
18	**Windows Key + L**	Lock your PC
19	**Windows Key + R**	Open the Run dialog box
20	**Windows Key + S**	Open Search
21	**Windows Key + I**	Open Settings
22	**Windows Key + X**	Open the Quick Link menu
23	**Windows Key + A**	Open Action Center
24	**Windows Key + V**	Open Clipboard history
25	**Windows + Shift + S**	Screenshot (Snip Tool)
26	**Windows Key + Print Screen**	Save screenshot file
27	**Windows Key + K**	Connect devices
28	**Windows Key + .**	Open emoji panel
29	**Ctrl + Shift + Esc**	Open Task Manager
30	**Ctrl + Alt + Del**	Open security screen
31	**Alt + F4**	Close active app
32	**Alt + Tab**	Switch between open apps
33	**Windows Key + Tab**	Open Task View
34	**F2**	Rename selected item
35	**F5**	Refresh active window
36	**Shift + Delete**	Permanently delete file
37	**Ctrl + Shift + N**	New folder
38	**Alt + Enter**	Open properties for selected item
39	**Windows Key + Ctrl + D**	New virtual desktop
40	**Windows Key + Ctrl + F4**	Close desktop

5. Company Logo Identification (UNSEEN)

- Participants should be able to identify logos from different companies (e.g. Windows, Microsoft, Coca-Cola, Nestlé etc).

6. Real vs AI (UNSEEN)

- Participants should be able to differentiate between real and AI pictures, sounds, graphics and text.

7. Canva Graphic Designing and Editing Skills

- **Canva Basics**
  Navigation · Creating designs · Choosing dimensions · Using templates · Managing pages
- **Text Editing**
  Adding text, changing fonts, sizes, colours, alignment, spacing and text effects
- **Elements & Graphics**
  Adding shapes, icons, stickers, illustrations, frames and grids
- **Image Editing**
  Uploading, cropping, resizing, applying filters, adjusting brightness and removing backgrounds
- **Design Principles**
  Colour theory, typography, alignment, spacing, contrast, visual hierarchy and balance
- **Advanced Editing**
  Layering, grouping, transparency, positioning, guides, animations and transitions
- **Project Designing**
  Creating posters, presentations, infographics, newsletters, brochures and social media posts
- **Exporting & Sharing**
  Downloading PNG, JPG and PDF files, sharing links and collaborating